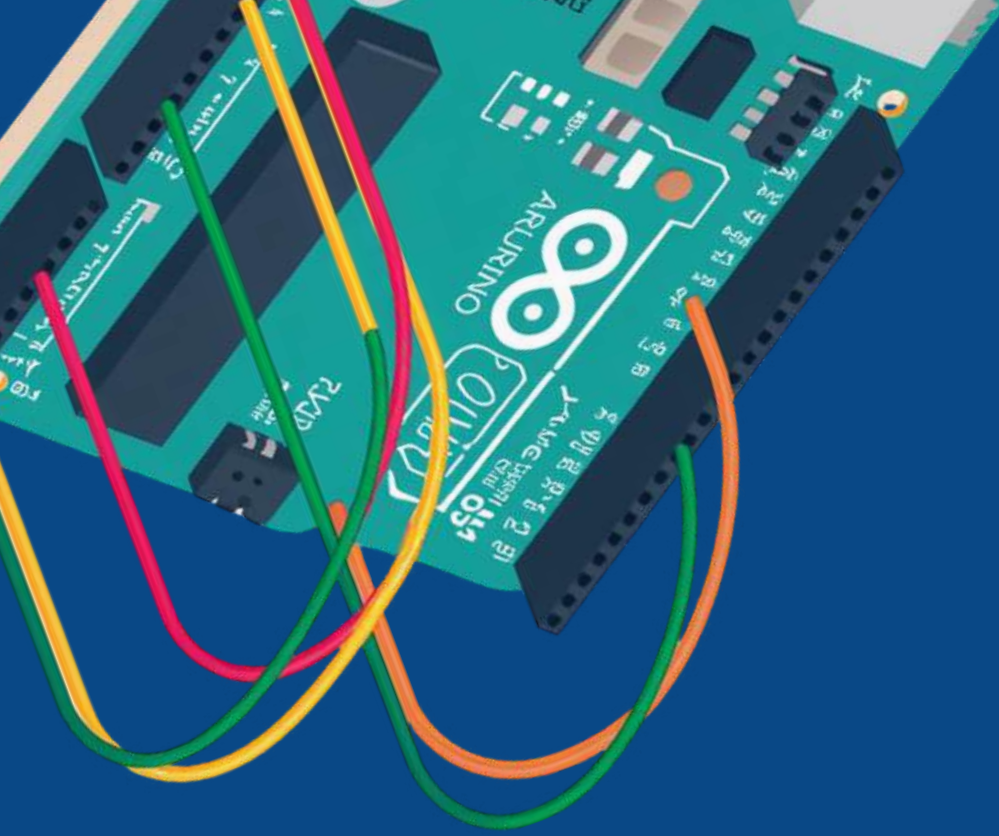




ARDUINO — ULTRASONIC DISTANCE SENSOR (HC-SR04)





HOW DO CARS KNOW WHEN TO STOP?

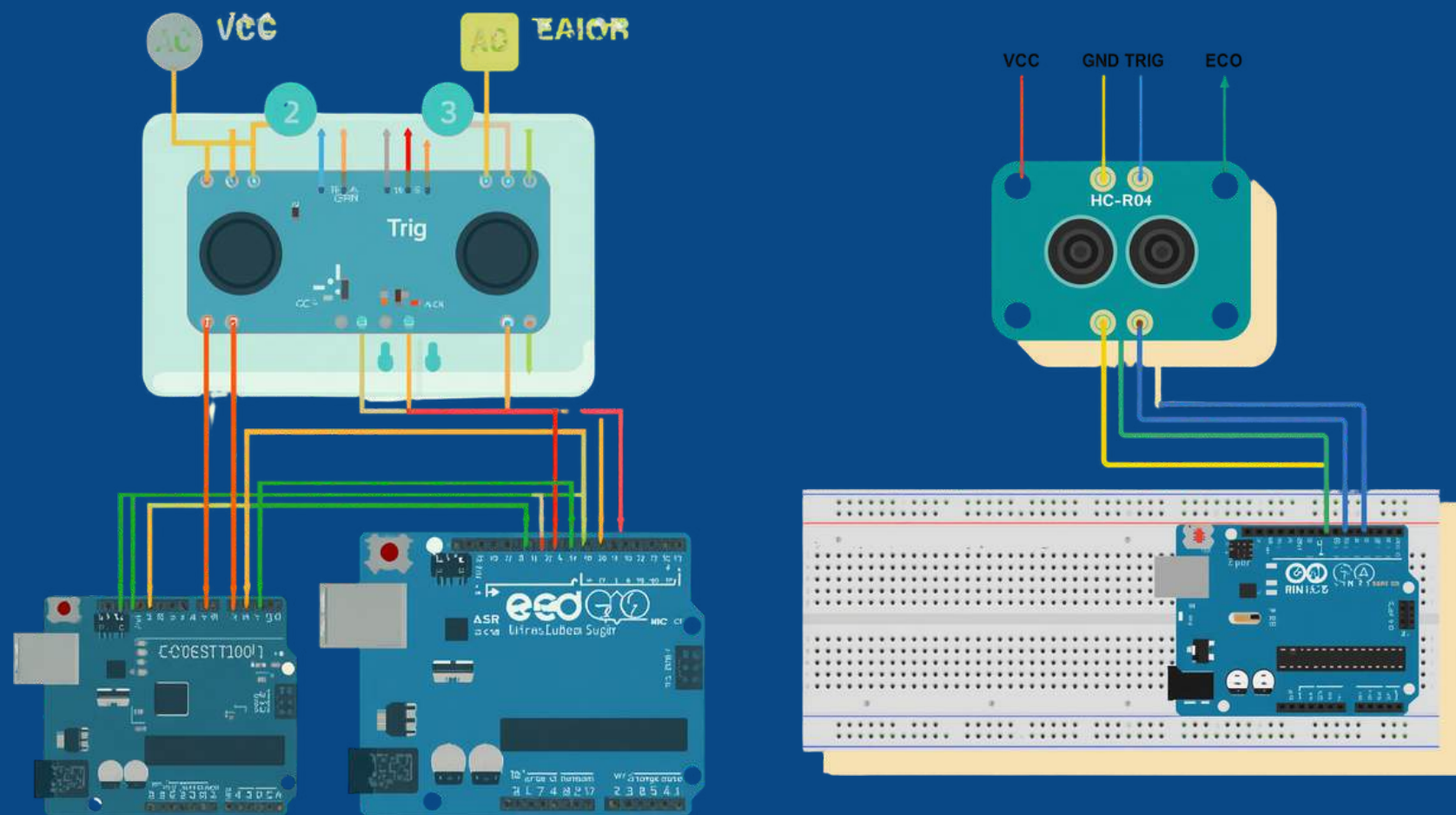
Have you ever seen a car stop just before hitting a wall while parking?

Rahul's uncle had to park his auto-rickshaw in a tight Bengaluru lane... but how did he avoid bumping?

 **Fun Fact:** Bats and dolphins use echolocation to 'see' with sound — just like the HC-SR04 sensor!



MEET THE HC-SR04 SENSOR!



The HC-SR04 has 4 important pins:

- ⚡ **VCC = Power (5V from Arduino)**
- ⏚ **GND = Ground**
- 📡 **Trig = Sends ultrasonic sound**
- 👂 **Echo = Listens for the bounce back**

We connect it to Arduino's 5V and GND to power it up!

HOW DOES IT MEASURE DISTANCE?

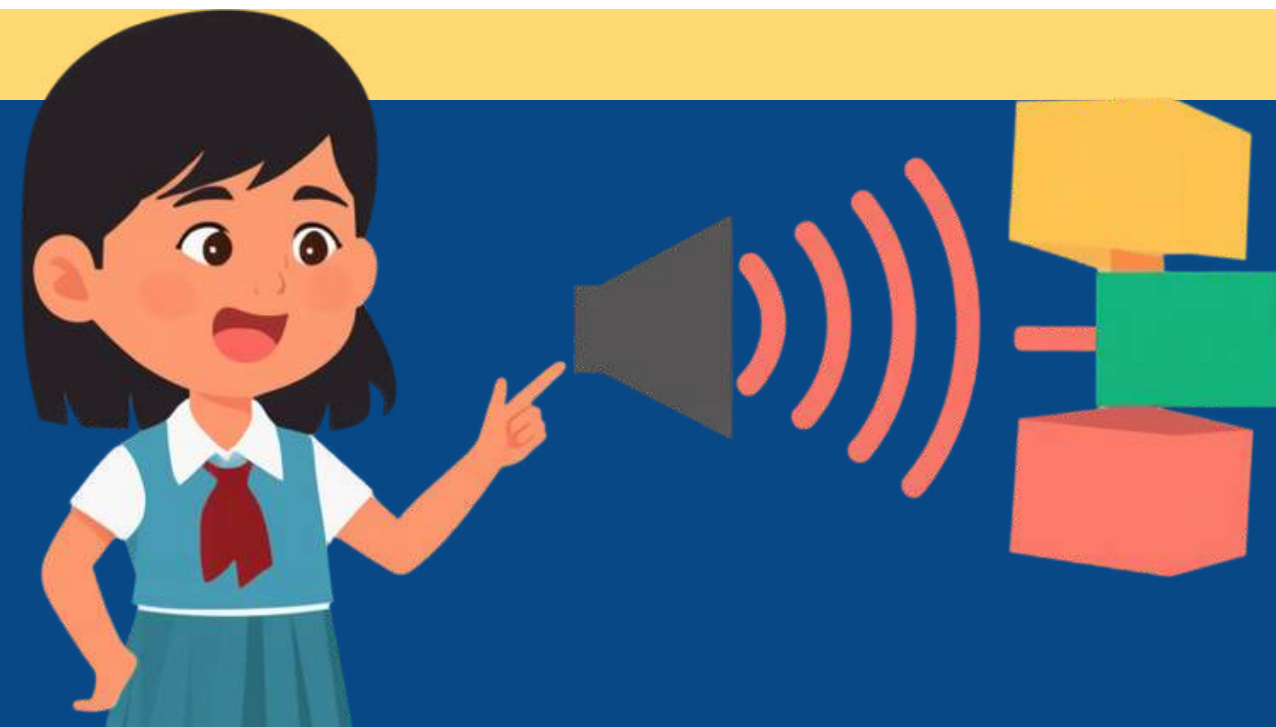


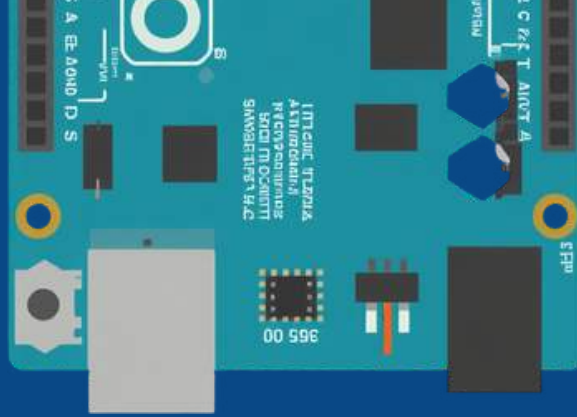
The sensor sends ultrasonic sound waves. They bounce back from objects!

 $\text{Distance} = (\text{Speed of Sound} \times \text{Time}) \div 2$

We divide by 2 because sound travels TO the object and back — that's TWO trips for one measurement!

 Speed of sound = 343 m/s in air





LET'S CODE!



Step 1: Set Up Pins

Set pin 2 as Trig (OUTPUT) and pin 3 as Echo (INPUT) in your setup() block to get ready!

Step 4: Print to Serial Monitor
Use Serial.begin(9600) and Serial.print(distance) to watch live readings on your computer screen.

Step 2: Send a Sound Pulse

Write LOW then HIGH on the Trig pin for 10 microseconds to send an ultrasonic burst.

Step 5: IF Distance < 15 cm
IF the object is closer than 15 cm → turn ON the red LED and buzzer on pin 7.
Alert activated! 🚨

Step 3: Read Distance in cm

Use pulseIn() on Echo pin to get travel time. Distance = (time × 0.034) ÷ 2. Print it to Serial Monitor!

Step 6: ELSE → Turn OFF
If nothing is close, turn OFF the LED and buzzer. Connect LED to pin 7 and buzzer to pin 8 on your Arduino!



QUICK QUIZ TIME!

Q1: Which pin sends the ultrasonic sound?
A) VCC B) Trig ✓ C) Echo

Q2: Why do we divide time by 2?
A) To get speed B) One-way distance ✓ C) Double the distance

Q3: What frequency range is ultrasonic sound?
A) Audible B) Below 20 Hz C) Above 20 kHz ✓

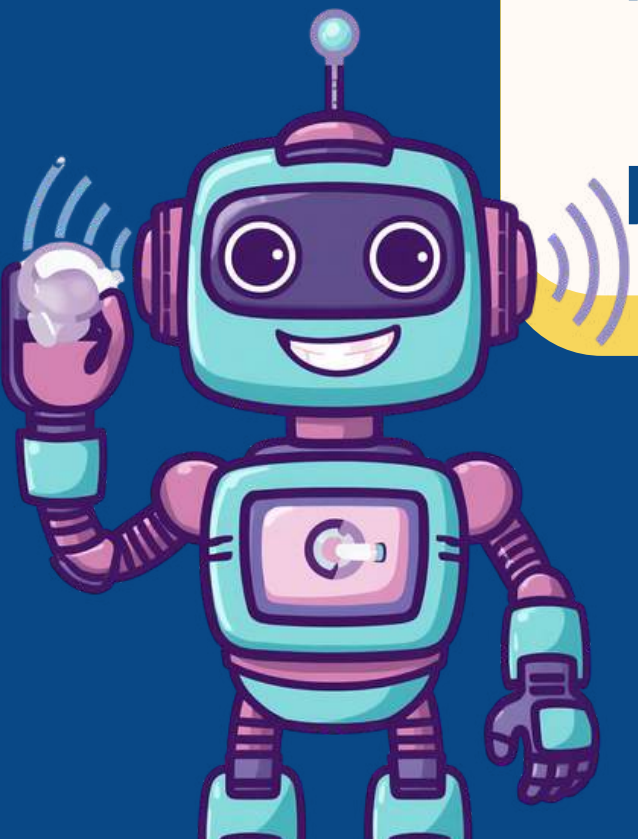
WRAP-UP: WHAT DID WE LEARN TODAY?

🦇 Echolocation helps animals AND sensors detect distance using sound waves!

📡 HC-SR04 has 4 pins: VCC (power), GND (ground), Trig (sends sound), Echo (listens back).

📐 $\text{Distance} = (\text{Speed of Sound} \times \text{Time}) \div 2$ — we divide by 2 because sound travels BOTH ways!

💡 Our block code reads the sensor and turns ON a red LED or buzzer when an object is closer than 15 cm!



MINI CHALLENGE!
DESIGN A MONSOON FLOOD WARNING
SYSTEM USING HC-SR04!
GREAT JOB TODAY — KEEP EXPLORING &
BUILDING! 🚀

